

K458

Element	Z	Mass Frac	Norm Mass	Atom Frac
Oxygen	8	0.3193	0.3193	0.629624
Silicon	14	0.2308	0.2308	0.259263
Zinc	30	0.0307	0.0307	0.0148077
Barium	56	0.4192	0.4192	0.0963056
--	14.504	1	1.0	1.0

A simple quantification not requiring references.

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Select one or more spectrum in the Spectrum List and then start the "Quantification Alien"

NIST DTSA-II

K458[Fri May 30 15:11:19 2014][0]

Counts

400
300
200
100
5 10
15 20

Spectrum Report
Default Detector
MIRA-3
SDD (Medium, 40
Spectrum List
K458[Fri May 30

Select a quantification mode

First page
Next: Specify the instrument

Select the mode which best describes the operation you wish to perform. The mode you select will determine what information you will be asked to provide and what information will be computed.

- ☒ Determine the composition of an 'unknown' spectrum by MLLSQ fitting to standards
- ☐ Determine the composition from k-ratios
- ☐ Determine the composition of an 'unknown' spectrum by fitting using a simplex method
- ☐ Estimate measured k-ratios from composition

Message: Select an analysis mode. More...

Back Next Finish Cancel

Detector orienta... [-0.819,-0.000,...
Detector position [27.851,0.000,-...
Detector thickness 0.45 mm
Detector type Silicon Drift Dete...
Detector window Metal: AD 2.2 f

Atomic Fraction

Silicon	0.2292±0.0043	0.2469±0.0046
Zinc	0.0322±0.0006	0.0149±0.0003
Barium	0.4344±0.0011	0.0957±0.0002

1 spectra selected.

NIST DTSA-II

File Process Tools Report Help

714 × 289

50 000

40 000

30 000

20 000

10 000

Counts

5 10

15 20

Spectrum Report

Default Detector

MIRA-3

SDD (Medium, 4096)

Spectrum List

K458[Fri May 30]

Quantification Alien

Previous: Select a quantification mode

Next: Specify standard spectra

Specify the instrument

Instrument

Acquired on the MIRA-3

Detector

using the SDD (Medium, 4096)

with calibration FWHM[Mn Ka] = 130.8 eV - 2014-02-18 00:00

Setting

at a beam energy of 20.0 keV.

Message:

More...

Back Next Finish Cancel

Detector orienta... [-0.819,-0.000,...

Detector position [27.851,0.000,-...

Detector thickness 0.45 mm

Detector type Silicon Drift Dete...

Detector window Model: AD 3.3 f

Silicon 0.2292±0.0043 0.2469±0.0046

Zinc 0.0322±0.0006 0.0149±0.0003

Barium 0.4344±0.0011 0.0957±0.0002

Atomic Fraction

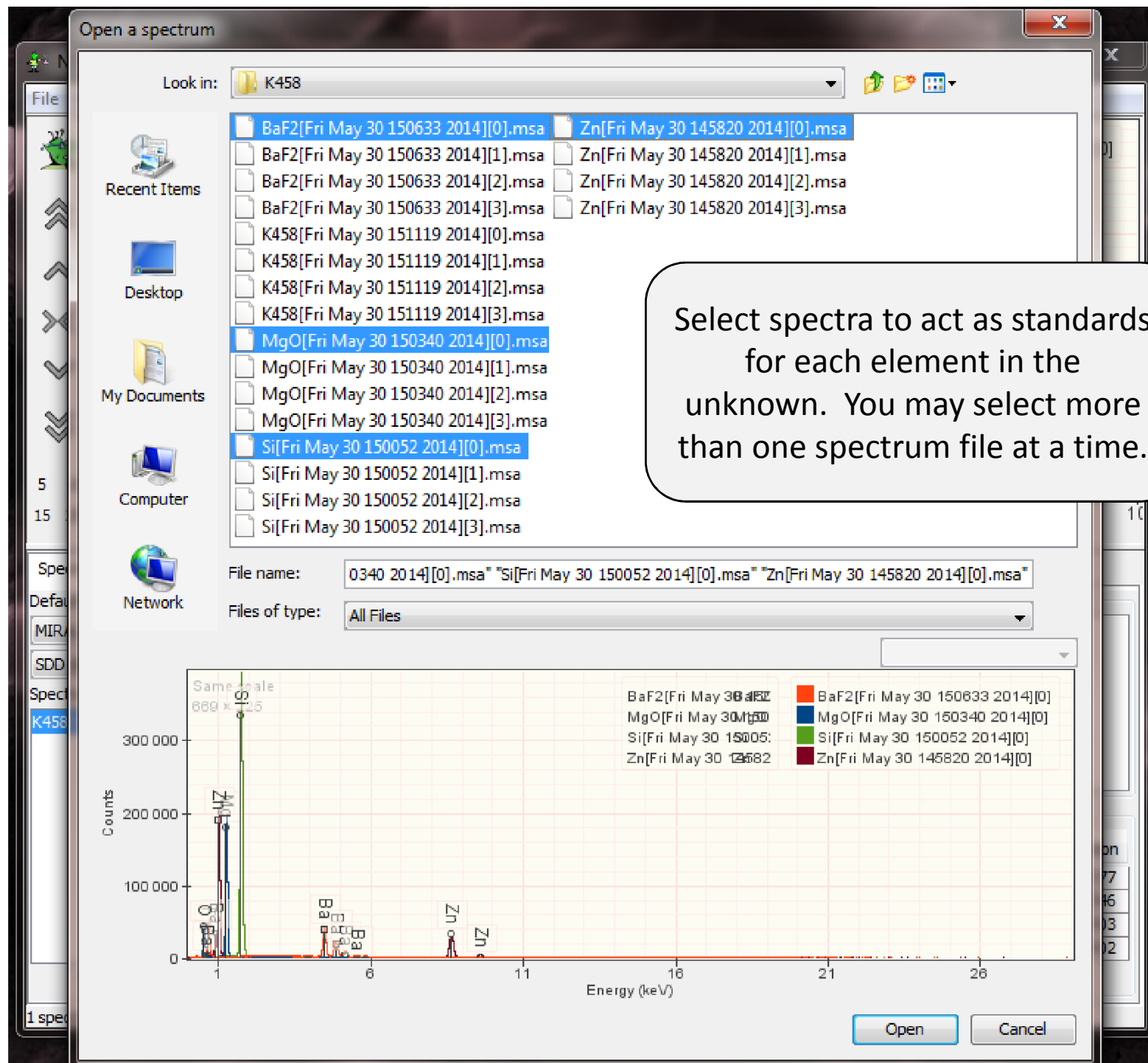
0.6426±0.0377

1 spectra selected.

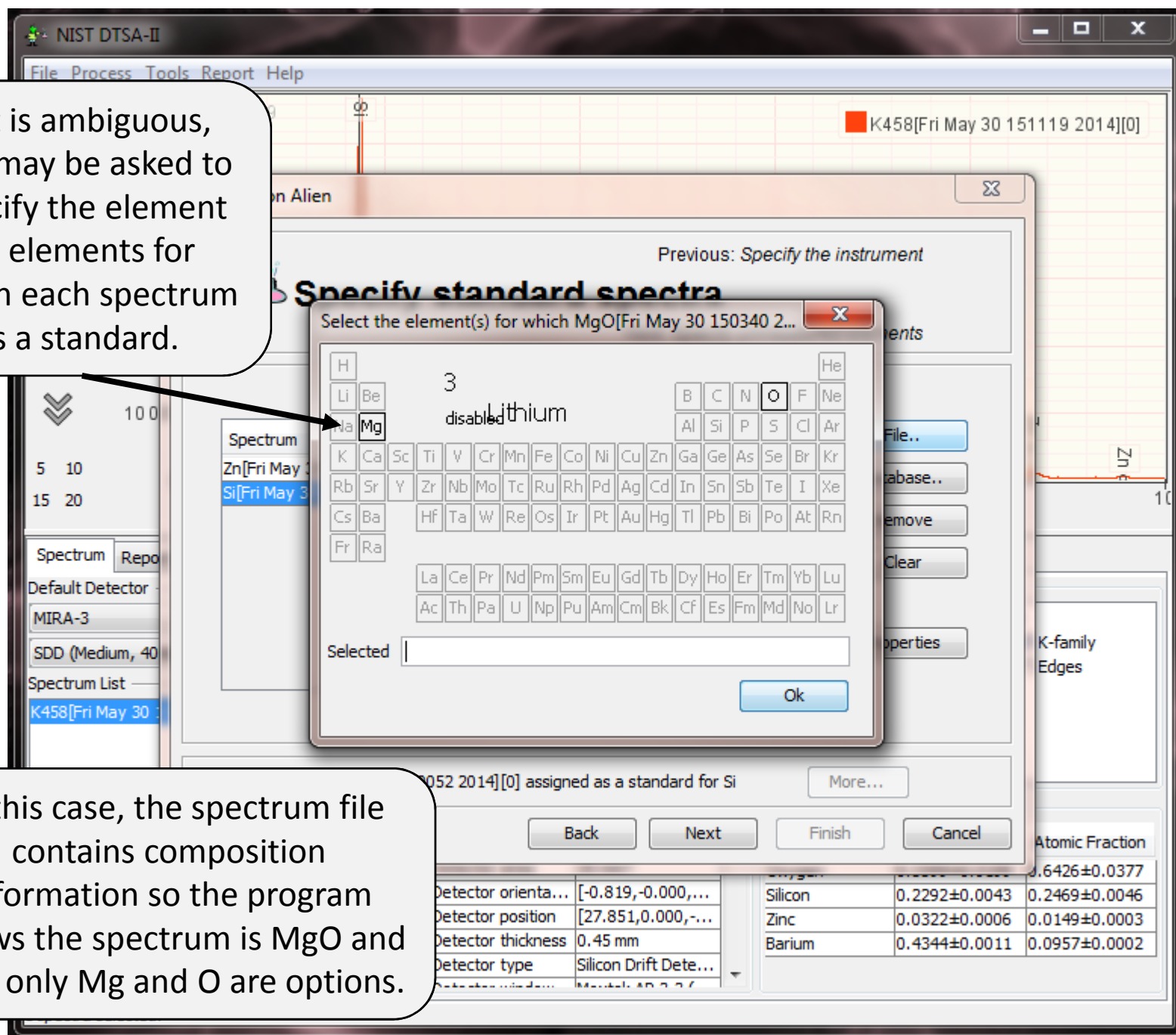
The information on this page is extracted from the spectrum and should be correct. Verify it and move on.

K-family Edges

Zn



If it is ambiguous,
you may be asked to
specify the element
or elements for
which each spectrum
is a standard.



In this case, the spectrum file
contains composition
information so the program
knows the spectrum is MgO and
thus only Mg and O are options.

NIST DTSA-II

File Process Tools Report Help

Previous: Spec

Next: Specify unmeasured elements

Specify standard spectra

Spectrum	Elements	Prob...	Live t...	Composition	Duane...
Zn[Fri May 30 1458...	Zn	0.990	56.9	Zn	19.920
Si[Fri May 30 15005...	Si	0.990	56.3	Si	20.012
MgO[Fri May 30 15...	O	0.990	57.8	MgO	19.639
BaF2[Fri May 30 15...	Ba	1.037	57.0	BaF2	19.991

File.. Database.. Remove Clear Properties

May 30 150633 2014][0] assigned as a standard for Ba More...

Back Next Finish Cancel

Detector orienta... [-0.819,-0.000,...
Detector position [27.851,0.000,-...
Detector thickness 0.45 mm
Detector type Silicon Drift Dete...
Detector window Model: AD 3.2.6

Silicon 0.2292±0.0043 0.2469±0.0046
Zinc 0.0322±0.0006 0.0149±0.0003
Barium 0.4344±0.0011 0.0957±0.0002

Atomic Fraction 0.6426±0.0377

K-family Edges

Counts 300 200 100

5 10 15 20

Spectrum Report

Default Detector MIRA-3

SDD (Medium, 40

Spectrum List

This panel summarizes the assignment of standards.

The Duane-Hunt should be close to the nominal beam energy.

Check that the probe current, live time and composition are correct. When the composition is not recorded in the spectrum file, the program will make a guess.

Sometimes one or more element may be estimated based on assumed chemistry. You can select assumptions here.

This time we will select "No extra element." In other words, all the elements in the unknown are accounted for by standards. ***You must account for every element in the unknown!***

K458[Fri May 30 15:11:19 2014][0]

Previous: Specify standard spectra
Next: Specify the reference spectra

Specify unmeasured elements

☒ No extra element
☐ Element by difference
☐ Oxygen by stoichiometry
☐ Waters of Crystallization

Element	Cation	Anion	As
Oxygen	1	-1	O
Zinc	1	1	ZnO
Barium	1	1	BaO
Silicon	1	2	SiO ₂

Specify how to handle unmeasured elements [More...](#)

[Back](#) [Next](#) [Finish](#) [Cancel](#)

Detector orienta...	[-0.819,-0.000,...]	Silicon	0.2292±0.0043	0.2469±0.0046
Detector position	[27.851,0.000,-...	Zinc	0.0322±0.0006	0.0149±0.0003
Detector thickness	0.45 mm	Barium	0.4344±0.0011	0.0957±0.0002
Detector type	Silicon Drift Dete...			
Detector window	Model: AD 2 2 6			

1 spectra selected.

K-family Edges

Atomic Fraction

0.6426±0.0377

DTSA-II checks the standards to ensure that no characteristic lines are obstructed. In this case, none are.

The standards can act as references so none of the “Regions-of-Interest” are listed as missing a reference spectrum.

Region-of-Interest	Spectrum	S/N
O All [0.368, 0.682 keV]	MgO[Fri May 30 150340 2...	Good 496
Zn L3-M5 + 6 others [0.75...	Zn[Fri May 30 145820 201...	Good 1285
Si All [1.558, 1.951 keV]	Si[Fri May 30 150052 2014...	Good 1829
Ba L3-M1 [3.819, 4.089 keV]	BaF2[Fri May 30 150633 2...	Poor 59
Ba L3-M5 + 5 others [4.25...	BaF2[Fri May 30 150633 2...	Good 730
Ba L2-N4 + 2 others [5.36...	BaF2[Fri May 30 150633 2...	Poor 93
Zn K α [8.363, 8.902 keV]	Zn[Fri May 30 145820 201...	Good 668
Zn K β [9.353, 9.791 keV]	Zn[Fri May 30 145820 201...	Ok 223

Some of the less intense line sets are listed as Poor (<100 S/N) or Ok signal-to-noise but at least one line per element is listed as good.

NIST DTSA-II

Specify the reference spectra

Previous: Specify the reference spectra

Next: Specify lines to quantify.

Message: Specify reference spectra (as necessary.)

Back Next Finish

1 spectra selected.

Detector orienta... [-0.819,-0.000,...]
Detector position [27.851,0.000,-...]
Detector thickness 0.45 mm
Detector type Silicon Drift Dete...
Detector window Model: AD 2.2.7

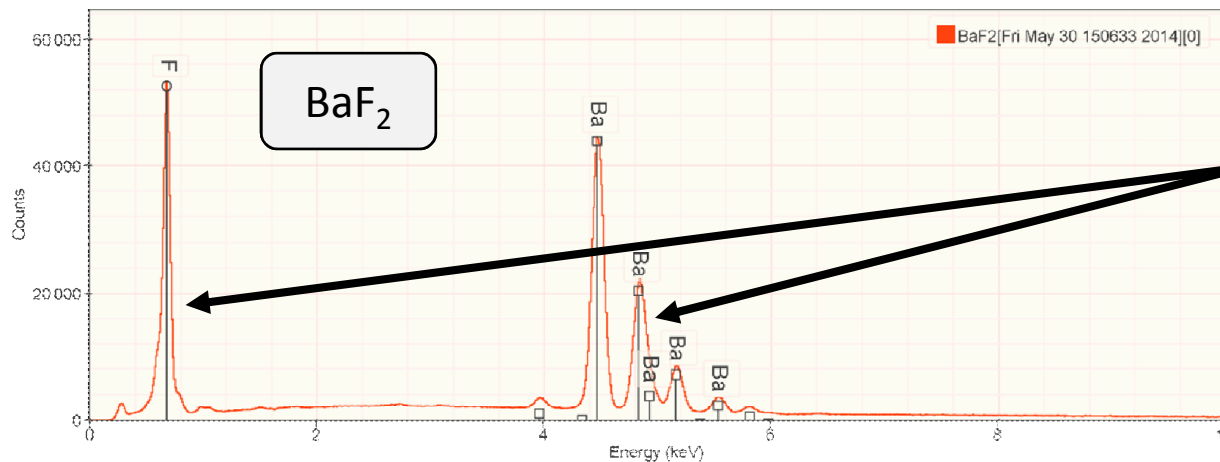
Silicon 0
Zinc 0
Barium 0

K-family Edges

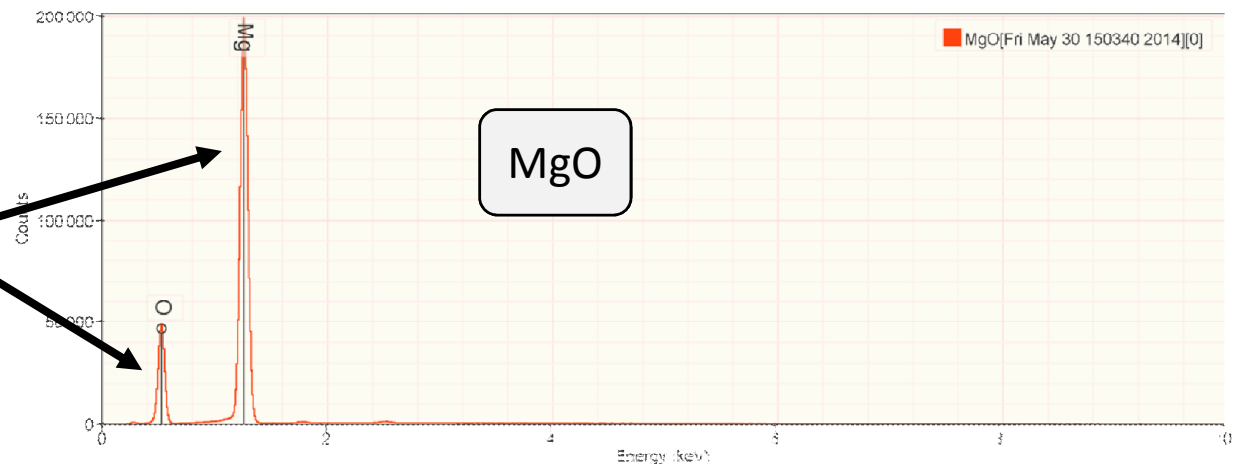
Spectrum Report
Default Detector MIRA-3
SDD (Medium, 40)
Spectrum List
K458[Fri May 30]

File...
Database...
Remove
Strip

What does “unobstructed” mean?



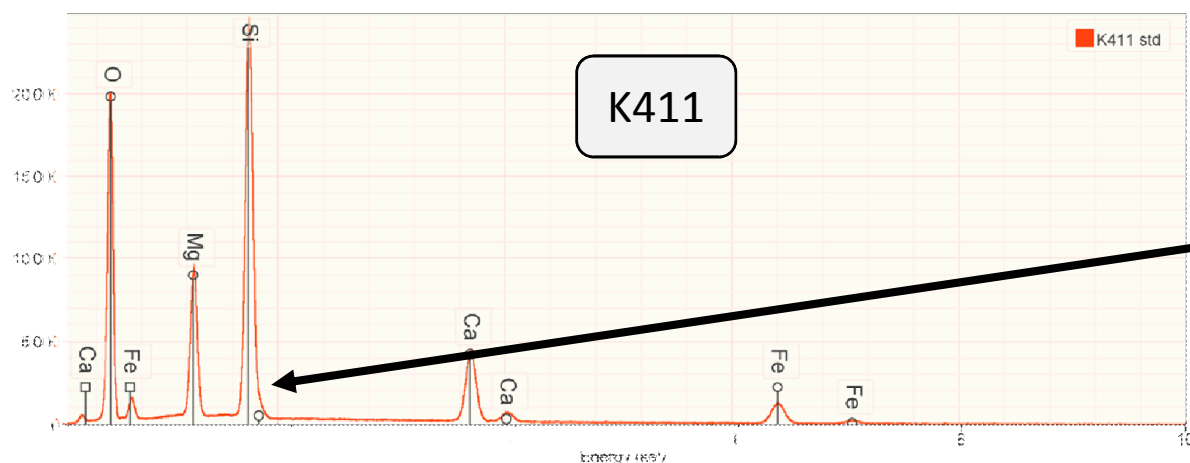
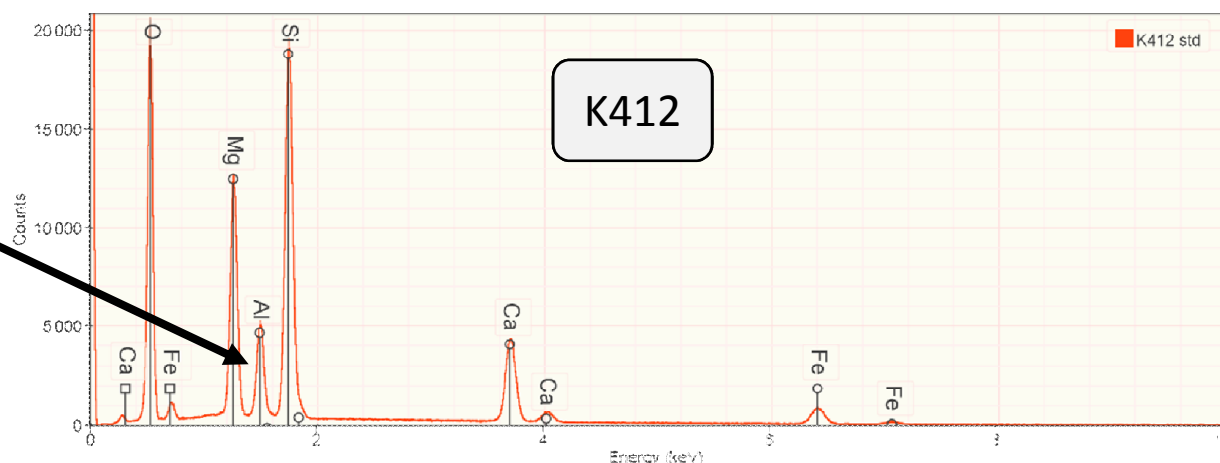
The Ba L and F K lines are separated so that they don't overlap.



The Mg K and O K are sufficiently separated, too.

An example of “obstructed”

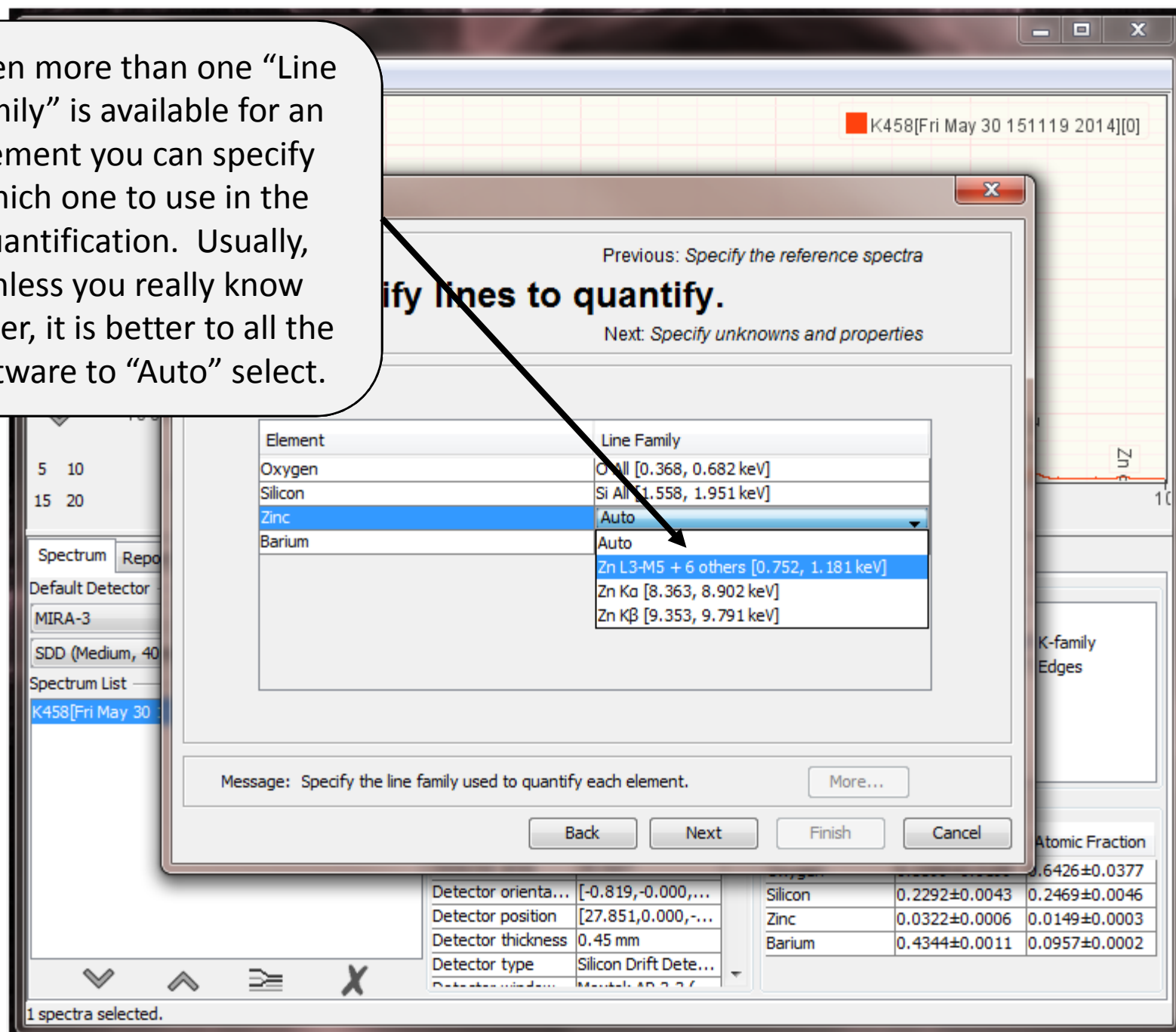
K412 could be suitable as a standard for Mg, Al and/or Si but Al is too close to Mg and Si. A reference is required for Al, Mg and Si.



On the other hand, K411 could be used as a standard for Mg and Si without any references because Mg and Si are sufficiently separated.

Let the software tell you whether references are needed.

When more than one “Line Family” is available for an element you can specify which one to use in the quantification. Usually, unless you really know better, it is better to all the software to “Auto” select.



NIST DTSA-II

File Process Tools Report

714 x 2

50 000

40 0

30 0

20 0

10 0

Counts

5 10

15 20

Spectrum Report

Default Detector

MIRA-3

SDD (Medium, 40

Spectrum List

K458[Fri May 20

Make sure that the "Live Time",
"Probe (nA)" are correct.

Quantification Area

Previous: Specify lines to quantify.

Specify unknowns and properties

Next: The results

Name	Live Time	Probe (nA)	Duane-Hunt	Shape
K458[Fri M...	57.9	1.037	19.973	Bulk

Add file

Remove

Properties

Sample Shape

More...

Finish

Cancel

Message: Specify the unknown

By default, the spectra
selected in "Spectrum List" are
displayed here.

K458[Fri May 30 151119 2014][0]

Zn

K-family Edges

Atomic Fraction

Carbon	0.6426±0.0377	
Silicon	0.2292±0.0043	0.2469±0.0046
Oxygen	0.0322±0.0006	0.0149±0.0003
Barium	0.4344±0.0011	0.0957±0.0002

1 spectra selected.

NIST DTSA-II

File Process Tools Report Help

714 × 289

Quantification Alien

Specify unknown

Duane-Hunt limit warning

The Duane-Hunt limit for the unknown spectrum is 19.973 keV.
The nominal beam energy is 20.000 keV.
The Duane-Hunt limit for the standard "MgO[Fri May 30 150340 2014][0]" differs by 0.361 keV from the beam energy.
A low Duane-Hunt limit suggests sample charging and can degrade the accuracy of the quantitative results.
Should we proceed?

Yes No

Message: Specify the unknown spectra

More...

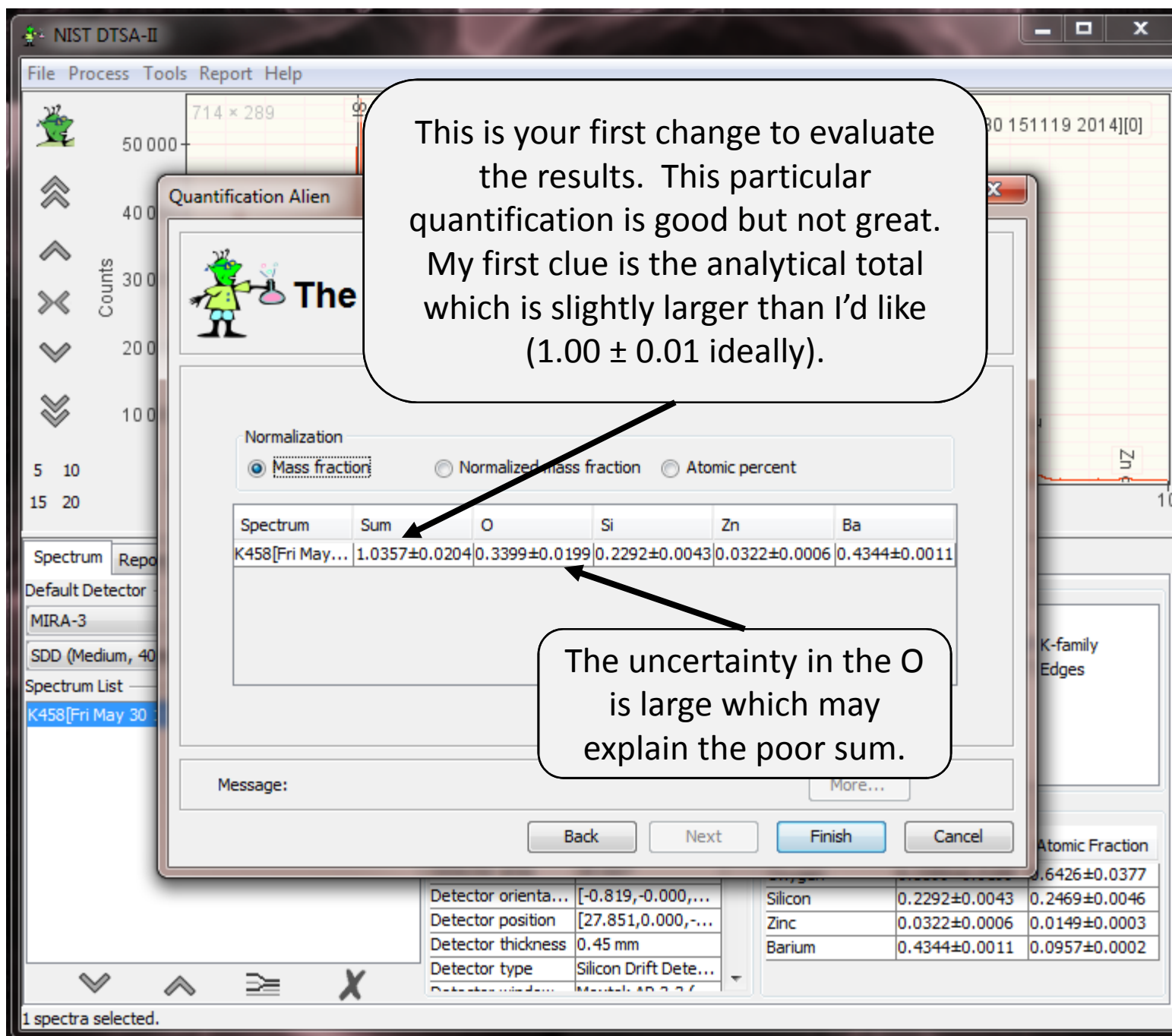
Back Next Finish Cancel

Detector orienta... [-0.8
Detector position [27.8
Detector thickness 0.45
Detector type Silico
Detector window

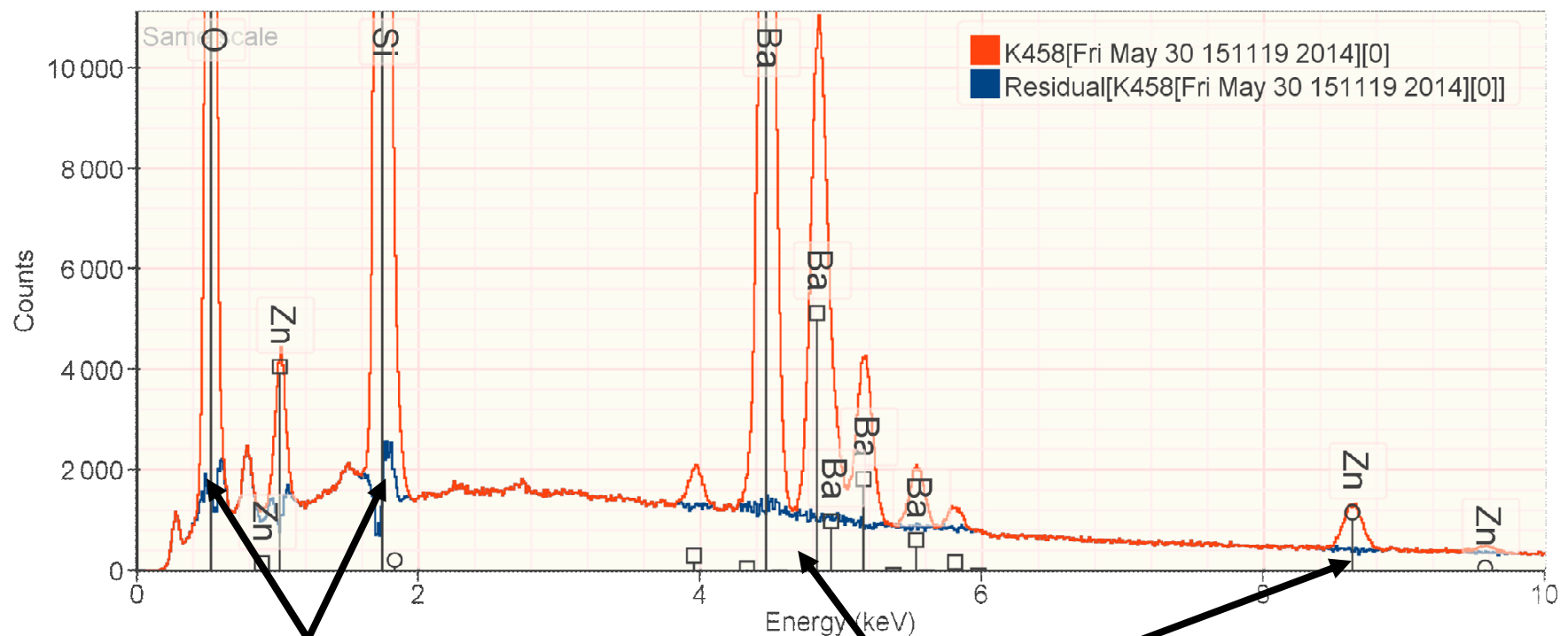
1 spectra selected.

You may get a warning about the "Duane-Hunt limit." The Duane-Hunt limit is an estimate of the actual energy with which the electron beam strikes the sample. It may be less than nominal if the sample is charging

You probably will want to select "Yes." However, if the results are poor, reconsider the warning.



After closing the “Quantification Alien,” the first thing I check is the residual. Did we account for all the elements? Does the shape of the continuum under the peaks look reasonable?



Less clean but still ok. Low energy lines typically have less clean residuals due to challenges fitting and chemical shifts.

Really clean residuals.

The matrix correction

Then I examine the report. The numbers in the boxes with the grey background are the suggested values to report.

Results

Spectrum	Quantity	O			Si			Zn			Ba			Sum
K458[Fri May 30 15:11:19 2014][0]	Line	O All			Si All			Zn Ka			Ba L3-M5 + 5 others			
Bulk	Z · A · F	1.142	0.835	0.998	1.098	0.837	1.002	0.927	0.963	1.000	0.851	1.015	1.000	
	k-ratios	0.8156 ±	0.0020		0.1355 ±	0.0002		0.0287 ±	0.0006		0.4792 ±	0.0012		
D-H = 19.973 keV	mass fraction	0.3399 ±	0.0007(I[std,O]) 0.0005(I[unk,O]) 0.0199([μ/ρ]) 0.0002(η) [0.0199]		0.2292 ±	0.0001(I[std,Si]) 0.0003(I[unk,Si]) 0.0043([μ/ρ]) 1.8·10 ⁻⁵ (η) [0.0043]		0.0322 ±	4.8·10 ⁻⁵ (I[std,Zn]) 0.0006(I[unk,Zn]) 6.6·10 ⁻⁵ ([μ/ρ]) 9.6·10 ⁻⁸ (η) [0.0006]		0.4344 ±	0.0006(I[std,Ba]) 0.0009(I[unk,Ba]) 0.0003([μ/ρ]) 7.0·10 ⁻⁶ (η) [0.0001]		1.0357
I = 1.037 nA	norm(mass fraction)	0.3282 ±	0.0192		0.2213 ±	0.0041		0.0311 ±	0.0006		0.4194 ±	0.0010		-
LT = 57.9 s	atomic fraction	0.0426 ±	0.0377		0.2469 ±	0.0046		0.0149 ±	0.0003		0.0957 ±	0.0002		
Residual	C:\Users\nritchie.NIST\AppData\Local\NIST\MSF\STDA\1504\1504 May-2014\residual5893831792160545253.msa													
Notes	Uncertainties are 1 σ and labeled by source. [unknown], [μ/ρ]: Absorption correction													

Alternative representations of the results.

An itemized uncertainty budget.

I happen to have a good idea what the correct answer is. Most of the error is in the O (expected) and the Ba (less expected.)

Element	Standard	Measured	Difference	% Difference
Si	23.08	22.92	-0.16	-0.7
O	31.93	33.99	2.06	6.5
Ba	41.92	43.44	1.52	3.6
Zn	3.07	3.22	0.15	4.9
4	100.00	103.57	2.57	8.9

Let's try again but this time we won't provide a standard for oxygen. Instead we will account for it by assumption.

NIST DTSA-II

K458[Fri May 30 151119 2014][0]

Previous: Specify the instrument

Specify standard spectra

Next: Specify unmeasured elements

Spectrum	Elements	Prob...	Live t...	Composition	Duane...
Zn[Fri May 30 1458...	Zn	0.990	56.9	Zn	19.920
Si[Fri May 30 15005...	Si	0.990	56.3	Si	20.012
BaF2[Fri May 30 15...	Ba	1.037	57.0	BaF2	19.991

File.. Database.. Remove Clear Properties

Message: BaF2[Fri May 30 150633 2014][0] assigned as a standard for Ba

Back Next Finish Cancel

Detector orienta... [-0.819,-0.000,...
 Detector position [27.851,0.000,-...
 Detector thickness 0.45 mm
 Detector type Silicon Drift Dete...
 Detector window Monte Carlo 2.2.6

	Atomic Fraction
Silicon	0.2292±0.0043 0.2469±0.0046
Zinc	0.0322±0.0006 0.0149±0.0003
Barium	0.4344±0.0011 0.0957±0.0002

1 spectra selected.

0 missing!

This time we will chose “Oxygen by stoichiometry” and these assumed valences (which are the default.)

NIST DTSA-II

K458[Fri May 30 151119 2014][0]

Previous: Specify standard spectra

Next: Specify the reference spectra

Specify unmeasured elements

☐ No extra element
☐ Element by difference
☒ Oxygen by stoichiometry
☐ Waters of Crystallization

Oxygen

Element	Cation	Anion	As
Zinc	1	1	ZnO
Barium	1	1	BaO
Silicon	1	2	SiO ₂

Message: Specify how to handle unmeasured elements

More...

Back Next Finish Cancel

1 spectra selected.

Detector orienta... [-0.819,-0.000,...]
 Detector position [27.851,0.000,-...]
 Detector thickness 0.45 mm
 Detector type Silicon Drift Dete...
 Detector window Model: AD 3.2.6

Atomic Fraction

Silicon	0.2292±0.0043	0.2469±0.0046
Zinc	0.0322±0.0006	0.0149±0.0003
Barium	0.4344±0.0011	0.0957±0.0002

Now the analytical total is within my desired tolerance.

The uncertainty in O is 5 times better.

Results

Spectrum	Quantity	O			Si			Zn			Ba			Sum
K458[Fri May 30 15:11:19 2014][0]	Line	O-by-stoichiometry			Si All			Zn Ka			Ba L3-M5 + 5 others			
Bulk	Z · A · F	-	-	-	1.193	0.544	1.002	0.932	0.963	1.000	0.855	1.017	1.000	
	k-ratios	-	-	-	0.1355	±	0.0002	0.0287	±	0.0006	0.4792	±	0.0012	
D-H = 19.973 keV	mass fraction	0.3149	±	0.0001(Ba) 0.0048(Si) 0.0002(η) [0.0048]	0.2254	±	0.0001(I[std,Si]) 0.0003(I[unk,Si]) 0.0042([μ/ρ]) 1.8·10 ⁻⁵ (η) [0.0042]	0.0320	±	4.8·10 ⁻⁵ (I[std,Zn]) 0.0006(I[unk,Zn]) 6.5·10 ⁻⁵ ([μ/ρ]) 9.5·10 ⁻⁸ (η) [0.0006]	0.4314	±	0.0006(I[std,Ba]) 0.0009(I[unk,Ba]) 0.0003([μ/ρ]) 6.9·10 ⁻⁶ (η) [0.0011]	1.0037
I = 1.037 nA	norm(mass fraction)	0.3137	±	0.0048	0.2245	±	0.0042	0.0319	±	0.0006	0.4298	±	0.0011	-
LT = 57.9 s	atomic fraction	0.6280	±	0.0095	0.2561	±	0.0048	0.0156	±	0.0003	0.1003	±	0.0002	
Residual	C:\Users\nritchie.NIST\AppData\Local\NIST\NIST DTSA-II Reports\2014\May\31-May-2014\residual891336416425454901.msa													
Notes	Uncertainties are 1 σ and labeled by source. (I[std unk]: Count statistics[standard unknown], [μ/ρ]: Absorption correction, η: Backscatter correction, []: combined)													

Overall better. Si is less accurate but O, Ba and Zn are all better. Not brilliant but ok.

Element	Standard	Measured	Difference	% Difference
Si	23.08	22.54	-0.54	-2.4
O	31.93	31.49	-0.44	-1.4
Ba	41.92	43.14	1.22	2.9
Zn	3.07	3.20	0.13	4.3
4	100.00	100.37	1.42	5.8

A tip for exporting the results

